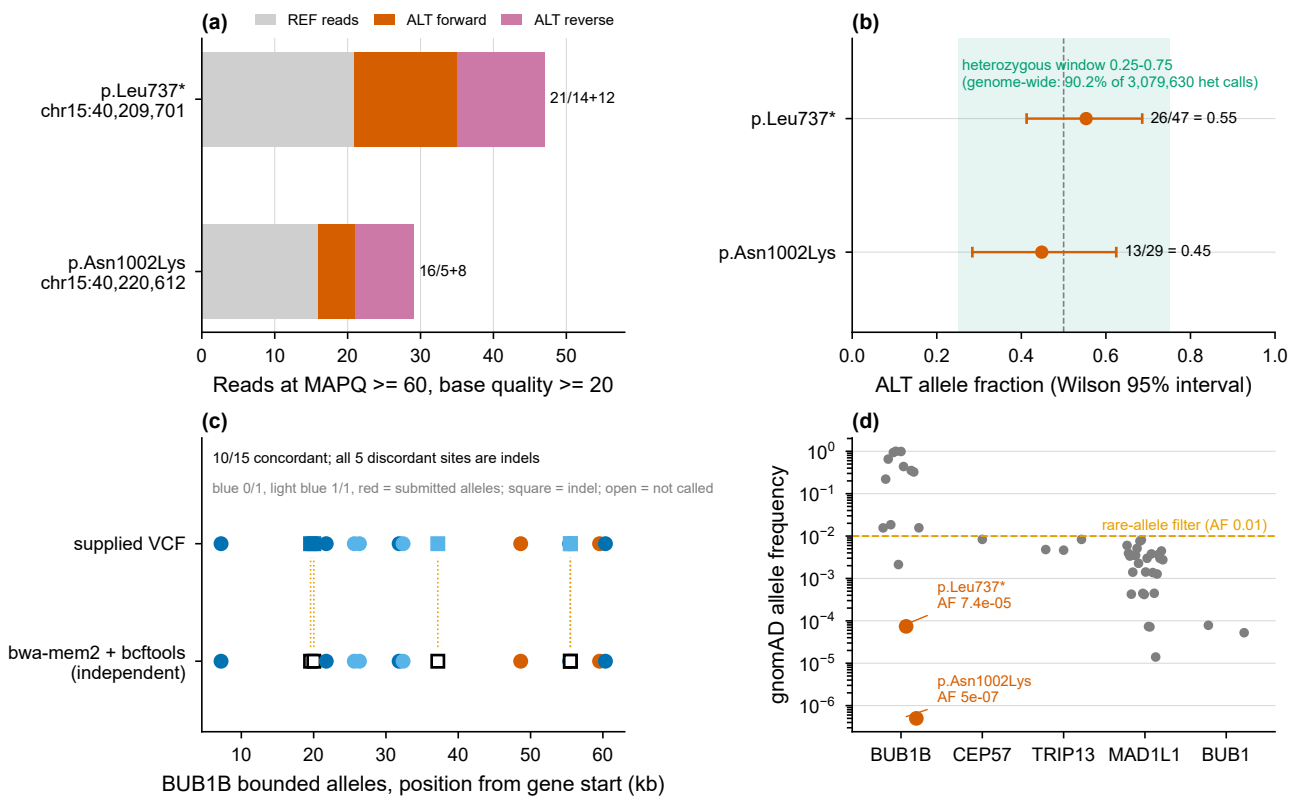

(a)
REF reads
ALT forward
ALT reverse
p.Leu737*
chr15:40,209,701
21/14+12
p.Asn1002Lys
chr15:40,220,612
16/5+8
0
10
20
30
40
50
Reads at MAPQ >= 60, base quality >= 20
(b)
heterozygous window 0.25-0.75
(genome-wide: 90.2% of 3,079,630 het calls)
p.Leu737*
26/47 = 0.55
p.Asn1002Lys
13/29 = 0.45
0.0
0.2
0.4
0.6
0.8
1.0
ALT allele fraction (Wilson 95% interval)
(c)
10/15 concordant; all 5 discordant sites are indels
blue 0/1, light blue 1/1, red = submitted alleles; square = indel; open = not called
supplied VCF
bwa-mem2 + bcftools
(independent)
10
20
30
40
50
60
BUB1B bounded alleles, position from gene start (kb)
(d)
gnomAD allele frequency
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
rare-allele filter (AF 0.01)
p.Leu737*
AF 7.4e-05
p.Asn1002Lys
AF 5e-07
BUB1B
CEP57
TRIP13
MAD1L1
BUB1